GRPO unraveled: A Small-Scale Empirical Study of RLVR
CMPT 981 Project Milestone

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Abstract

[TODO: Abstract. Write last: the project goal in one sentence, the three papers reviewed, and the progress made since the proposal.]

1 Introduction

[TODO: Introduction]

2 Paper Reviews

[TODO: Shared notation (a few lines)]

2.1 Success amplification [Mroueh, 2025]

Setting and assumptions. [TODO: Setting and assumptions]

Main results. [TODO: Main results]

Connection to our project. [TODO: Connection to our project]

Limitations and open questions. [TODO: Limitations and open questions]

2.2 Gradient gap and step-size thresholds [Suk and Duan, 2025]

Setting and assumptions. [TODO: Setting and assumptions]

Main results. [TODO: Main results]

Connection to our project. [TODO: Connection to our project]

Limitations and open questions. [TODO: Limitations and open questions]

2.3 Pass@k advantage shaping [Thrampoulidis et al., 2026]

Setting and assumptions. [TODO: Setting and assumptions]

Main results. [TODO: Main results]

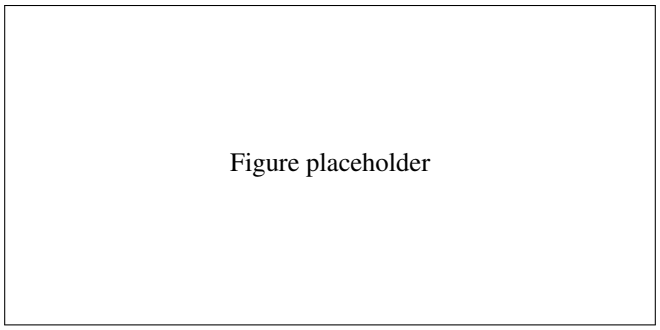


Figure 1: [TODO: Caption]

Connection to our project. [TODO: Connection to our project]

Limitations and open questions. [TODO: Limitations and open questions]

3 Progress

[TODO: Progress since the proposal]

4 Updated Plan

[TODO: Remaining work and timeline]

5 Contributions

- **Ankush Singh:** review of Suk and Duan [2025]. [TODO: other work]
- **Simon Doan:** review of Thrampoulidis et al. [2026]. [TODO: other work]
- **Azaan Rehman:** review of Mroueh [2025]. [TODO: other work]

References

Youssef Mroueh. Reinforcement learning with verifiable rewards: GRPO's effective loss, dynamics, and success amplification. *arXiv preprint arXiv:2503.06639*, 2025.

Joe Suk and Yaqi Duan. On the optimization dynamics of RLVR: Gradient gap and step size thresholds. *arXiv preprint arXiv:2510.08539*, 2025.

Christos Thrampoulidis, Sadegh Mahdavi, and Wenlong Deng. Advantage shaping as surrogate reward maximization: Unifying pass@K policy gradients. *Transactions on Machine Learning Research*, 2026. arXiv:2510.23049.